

Advances in Deep Transformer Architectures for NLP

Abstract

This paper introduces the architecture of modern deep neural network models, focusing primarily on transformer-based language representations. We analyze the math underlying multi-head self-attention mechanisms and how they optimize loss functions to capture token dependencies.

Methodology & Deep Learning Architectures

Our methodology utilizes a decoder-only transformer block configured with 32 attention heads, mapping tokens into a 4096-dimensional latent space. During training, we use an Adam optimizer with a cosine learning rate scheduler to reduce cross-entropy loss over a massive dataset of text abstracts.

Experimental Results & Accuracy

We evaluate the model on standard language benchmarks. The results demonstrate that increasing parameter capacity yields zero-shot accuracy improvements. Fine-tuning adaptation achieves state-of-the-art accuracy levels on complex question answering benchmarks.

Conclusion & Future Work

We have shown that large-scale model training enhances generalization. Future work will investigate model pruning and distillation techniques to optimize real-time inference latency.